

TIANHE, China

WMO: 574940

Lat: 30.5977N

Lon: 114.0506E

Elev: 82

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.5	30.0	10.7	9.5	38.9	16.0	12.4	38.9	15.7	38.7	14.1	39.1	4.5	20	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	97.2	81.7	95.3	81.2	93.4	80.6	84.8	92.6	83.6	91.2	82.5	89.9	8.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.7	171.2	89.5	82.0	167.3	88.9	80.6	159.4	87.5	49.5	92.2	47.9	91.0	46.6	90.3	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	13.5	11.6	23.7	100.1	3.3	2.3	21.3	101.8	19.4	103.1	17.6	104.4	15.2	106.1
			21.2	86.5	2.7	1.7	19.3	87.8	17.7	88.8	16.2	89.7	14.3	91.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	39.9	44.8	53.7	64.5	73.3	79.6	85.3	84.0	76.4	66.1	54.6	43.9	
	DBStd	16.66	5.64	7.11	7.63	7.08	5.52	4.88	4.96	5.29	5.77	6.01	7.07	5.55	
	HDD50	770	316	173	46	2	0	0	0	0	0	34	199		
	HDD65	2826	777	566	357	91	5	0	0	2	58	317	653		
	CDD50	5861	3	28	162	438	721	887	1094	1053	791	500	173	11	
	CDD65	2442	0	1	8	77	261	437	629	588	343	93	5	0	
	CDH74	26125	0	3	56	530	2023	4395	8271	7293	2987	540	27	0	
	CDH80	11805	0	1	9	126	653	1733	4387	3648	1129	117	2	0	
Wind	WSAvg	5.2	4.8	5.2	5.5	5.6	5.2	5.1	5.9	5.6	5.2	4.6	4.6	4.8	
Precipitation	PrecAvg	48.0	1.3	2.3	3.8	5.2	5.9	8.3	5.9	5.1	3.3	3.4	2.3	1.1	
	PrecMax	74.6	2.8	7.2	8.6	10.1	11.9	17.8	21.0	19.1	8.7	16.1	6.6	4.2	
	PrecMin	28.8	0.0	0.1	0.7	1.5	1.4	0.5	1.1	0.0	0.1	0.0	0.0	0.0	
	PrecStd	11.7	0.7	1.5	1.7	2.1	2.5	4.4	4.3	5.4	2.2	3.3	1.6	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.7	69.7	80.0	88.0	92.9	96.4	100.2	100.4	95.4	88.0	77.3	64.4	
		MCWB	49.5	59.7	67.2	71.6	72.9	80.1	82.6	81.0	79.3	72.0	62.5	51.9	
	2%	DB	56.1	64.1	73.8	83.7	89.5	93.3	98.0	97.2	91.8	82.8	73.0	59.4	
		MCWB	47.2	53.8	61.7	68.8	72.9	79.4	82.0	81.9	77.5	68.6	60.5	49.6	
	5%	DB	52.4	59.4	70.0	80.3	86.4	91.1	96.2	95.2	89.3	79.9	69.1	56.4	
		MCWB	45.3	50.8	59.5	66.9	72.2	78.2	81.7	81.8	76.5	66.5	58.7	47.7	
	10%	DB	49.4	55.7	66.2	76.6	83.8	88.3	94.1	93.0	86.3	76.7	65.8	53.4	
		MCWB	43.4	48.8	56.8	65.4	71.3	77.0	81.3	81.0	74.8	65.3	56.9	46.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	61.4	69.5	75.2	79.1	84.1	86.5	86.6	83.3	75.2	66.6	55.1
			MCDB	57.0	67.9	76.8	83.7	86.9	91.7	94.3	95.0	90.5	81.5	71.9	60.5
2%		WB	49.2	55.6	64.3	71.8	76.5	82.0	84.9	85.0	81.0	71.7	63.2	51.6	
		MCDB	53.7	60.6	71.6	79.6	83.7	88.6	92.9	92.8	87.9	79.2	69.1	56.9	
5%		WB	46.5	52.2	61.2	69.4	74.9	80.6	83.7	83.7	79.4	69.5	60.6	49.4	
		MCDB	51.2	57.5	67.8	76.9	82.0	87.3	91.6	91.1	86.2	76.4	66.5	54.3	
10%		WB	43.8	49.9	58.4	67.2	73.3	79.2	82.6	82.5	77.6	67.7	58.5	47.2	
		MCDB	48.2	54.5	64.5	73.8	80.2	85.6	90.5	89.5	83.8	73.7	63.9	51.9	
Mean Daily Temperature Range		5% DB	MDBR	12.4	13.0	14.5	15.1	14.7	13.0	12.6	13.1	14.2	14.6	14.5	13.7
			MCDBR	19.8	20.3	21.0	20.7	19.5	16.5	14.9	15.9	17.5	20.3	20.3	20.3
	MCWBR		11.7	11.4	11.0	9.6	8.2	7.1	6.2	6.4	7.3	8.4	9.7	11.3	
	5% WB	MCDBR	17.1	16.9	18.5	17.6	16.1	14.6	14.4	14.8	15.3	16.7	16.5	17.6	
		MCWBR	11.1	10.6	11.0	9.9	8.3	7.7	7.1	7.2	8.1	8.2	8.9	10.6	
Clear-Sky Solar Irradiance	taub	0.666	0.720	0.745	0.741	0.719	0.728	0.664	0.702	0.726	0.722	0.686	0.653		
	taud	1.531	1.469	1.452	1.500	1.578	1.587	1.738	1.640	1.570	1.525	1.554	1.572		
	Ebn at Noon	172	179	189	198	204	202	214	204	191	176	166	166		
	Edh at Noon	75	88	95	93	87	86	73	80	83	82	73	69		
All-Sky Solar Radiation	RadAvg	642	730	1062	1278	1379	1388	1598	1484	1241	971	780	647		
	RadStd	110	185	79	140	135	144	164	156	136	131	98	122		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	-1.24	N/A	+1.01	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page